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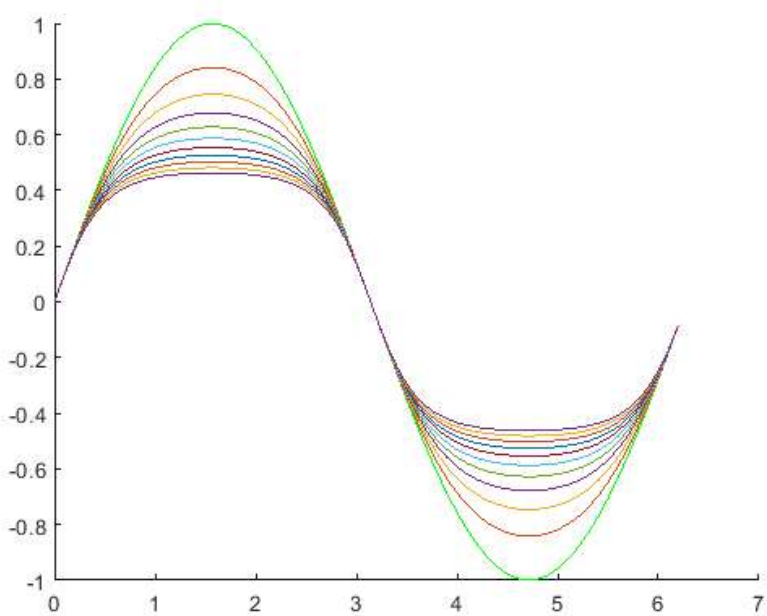
Exercise 1 %% % Adding integers from 1 to 10 with a for loop %

```
x = 0;
for i = 1:1:10      % iterate over all numbers from 1 to 10
    x = x + i;      % add them one by one to x
end

% Print to the console
fprintf('Exercise 1:\nSummation of all integers from 1 to 10 is: %.f \n\n', x)
```

```
Exercise 1:
Summation of all integers from 1 to 10 is: 55
```

Exercise 2 %% % Plotting iterations of y = sin(x) %

```
vector = 0:0.1:2*pi;    % Define the vector
y = sin(vector);        % create the starting value
figure(1)
hold on
plot(vector, y, 'g')    % plot first curve
for i = 1:1:10
    y = sin(y);          % now iterate
    plot(vector, y)
end
hold off
```



Exercise 3 %% % Using if/else to display passing or non-passing grades %

```
%grade = input('Exercise 3:\nEnter your grade for the class: ');
grade = 0.9

% If the grade given/set is greater than or equal to 90%, print message to the console
if (grade >= 0.9)
   fprintf('You aced the course.\n\n')

% If the grade given/set is les than 90% and greater than or equal to 0.8, print message to the console
elseif ((0.9 > grade) && (grade >= 0.8))
        fprintf('You almost aced the course.\n\n')

% If the previous two don't apply, you obviously didn't ace the course \/ \/ \/
else
    fprintf("You didn't ace the course...nice try.\n\n")

end
```

```
grade =

    0.9000

You aced the course.
```

Exercise 4 %% % Using a while loop to divide a large number %

```
num = 1000;
count = 0;
while (num >= 1)         % divide num by two until num becomes less than 1
    num = num/2;
    count = count + 1;  % Add 1 to the counter every time we divide by 2, until we can't anymore. We display count later
end
fprintf('Exercise 4:\nTotal number of iterations to achieve a number less than 1 from dividing 1000 by 2: %.f \n\n', count)
```

```
Exercise 4:
Total number of iterations to achieve a number less than 1 from dividing 1000 by 2: 10
```

Exercise 5 %% % Displaying arccosine and arcsine of various angles %

```
%x = input('Exercise 5:\nEnter a number between -1 and 1: ');     % Ask the user for input between -1 and 1
x = 0.5

if ((x <= -1) || (x >= 1))
    error("ERROR: Enter a valid number\n")          % If the input isn't in this range, produce an error
end

% Calculate arc cos and sin in radians
acos_rad = acos(x);
asin_rad = asin(x);

% Add conversion to degrees
acos_deg = (180/pi)*acos_rad;
asin_deg = (180/pi)*asin_rad;

% Now for the printing (I do 4 prints to make it a bit more readable)
fprintf("arccos(%g) = %.2f radians\n", x, acos_rad)
fprintf("arccos(%g) = %.2f degrees\n", x, acos_deg)
fprintf("arcsin(%g) = %.2f radians\n", x, asin_rad)
fprintf("arcsin(%g) = %.2f degrees\n", x, asin_deg)
```

```
x =

    0.5000
```

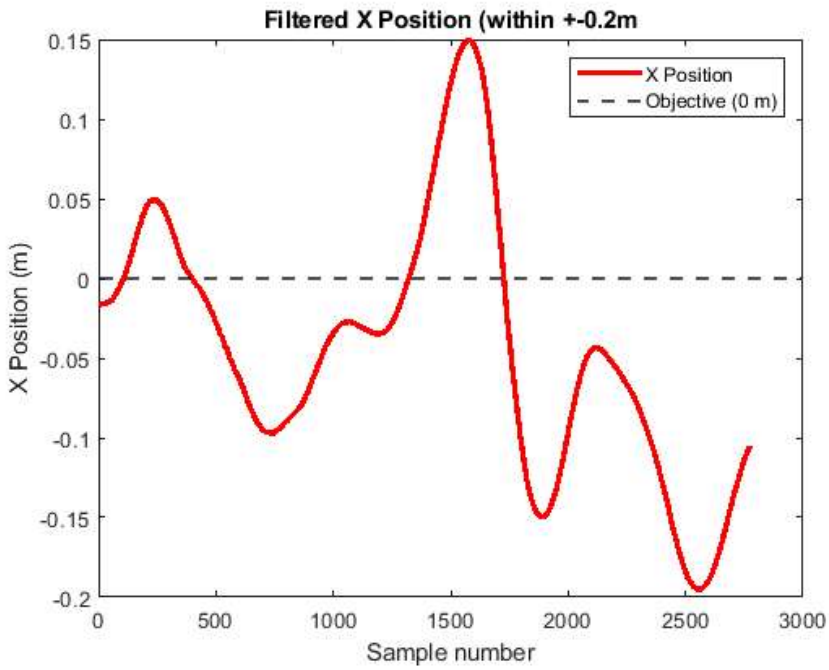
```
arccos(0.5) = 1.05 radians
arccos(0.5) = 60.00 degrees
arcsin(0.5) = 0.52 radians
arcsin(0.5) = 30.00 degrees
```

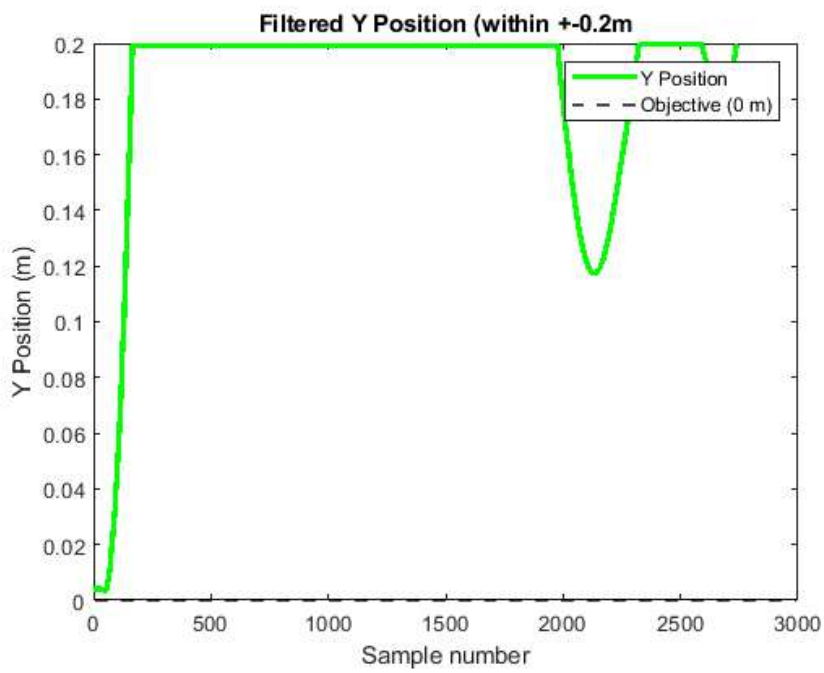
Exercise 6 %% % Digesting drone data and sorting it to plot proximity events %

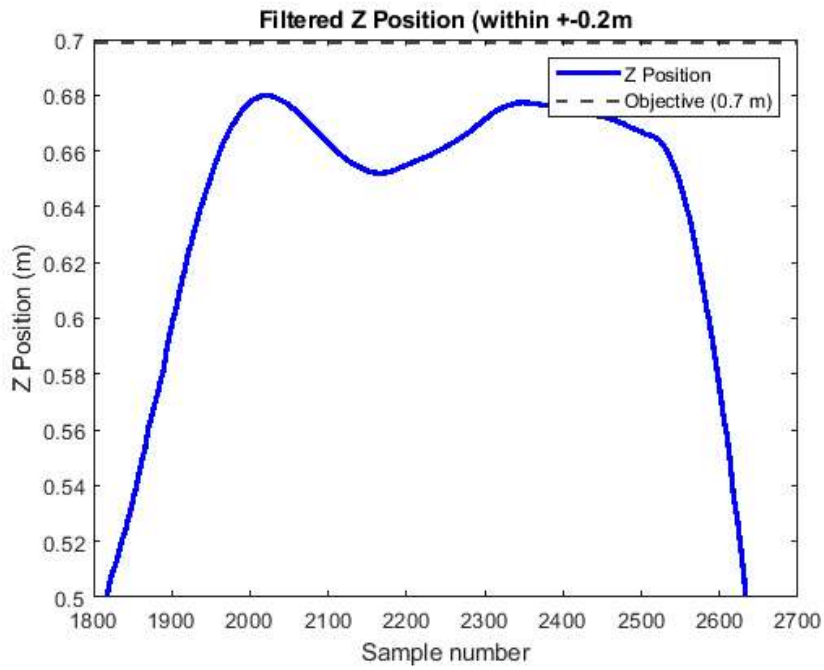
```
load('Lab2_Exercise6.mat')     % Load the drone data px, py, pz

% Filter the matricies to be only values within 0.2m of the target value
x_filtered = abs(p_x - 0.0) <= 0.2;
y_filtered = abs(p_y - 0.0) <= 0.2;
z_filtered = abs(p_z - 0.7) <= 0.2;

% Create time vector
T = 1:length(p_x);

% ---- X Plot ------

% Plot T and p_x values that match up with filterd x values, with a red line
figure(2)
plot(T(x_filtered), p_x(x_filtered), 'r', 'LineWidth', 2.5, 'DisplayName', "X Position")

% Create an "Objective" line, x and y label, and a title for the graph
hold on
yline(0, 'k--', 'LineWidth', 1.5, 'DisplayName', 'Objective (0 m)')
xlabel('Sample number')
ylabel('X Position (m)')
title('Filtered X Position (within +-0.2m')
legend
hold off

% ---- Y Plot ------

% Plot T and p_y values that match up with filterd y values, with a green line
figure(3)
plot(T(y_filtered), p_y(y_filtered), 'g', 'LineWidth', 2.5, 'DisplayName', "Y Position")

% Create an "Objective" line, x and y label, and a title for the graph
hold on
yline(0, 'k--', 'LineWidth', 1.5, 'DisplayName', 'Objective (0 m)')
xlabel('Sample number')
ylabel('Y Position (m)')
title('Filtered Y Position (within +-0.2m')
legend
hold off

% ---- Z Plot ------

% Plot T and p_z values that match up with filterd z values, with a blue line
figure(4)
plot(T(z_filtered), p_z(z_filtered), 'b', 'LineWidth', 2.5, 'DisplayName', "Z Position")

% Create an "Objective" line, this time at 0.7m, x and y label, and a title for the graph
hold on
yline(0.7, 'k--', 'LineWidth', 1.5, 'DisplayName', 'Objective (0.7 m)')
xlabel('Sample number')
ylabel('Z Position (m)')
title('Filtered Z Position (within +-0.2m')
legend
hold off

% ----- Combined Plot ------

% Plot all the lines together with different colors
figure(5)
hold on
plot(T(x_filtered), p_x(x_filtered), 'r', 'LineWidth', 2.5, 'DisplayName', "X Position")
plot(T(y_filtered), p_y(y_filtered), 'g', 'LineWidth', 2.5, 'DisplayName', "Y Position")
```

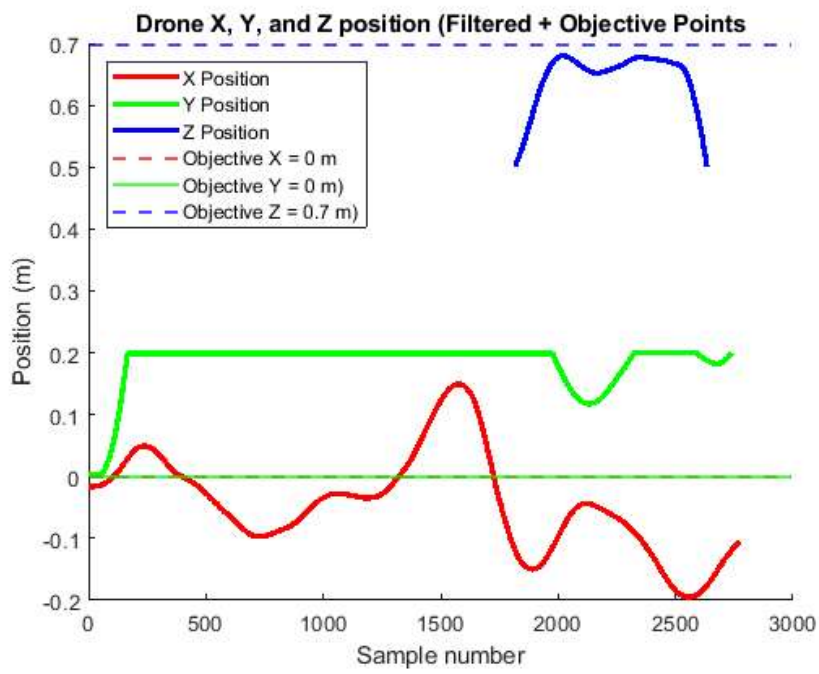
```
plot(T(z_filtered), p_z(z_filtered), 'b', 'LineWidth', 2.5, 'DisplayName', "Z Position")

% Create the three "Objective" lines, x and y label, and the title, as well as the legend
yline(0, 'r--', 'LineWidth', 1.5, 'DisplayName', 'Objective X = 0 m')
yline(0, 'g-', 'LineWidth', 1.5, 'DisplayName', 'Objective Y = 0 m)')
yline(0.7, 'b--', 'LineWidth', 1.5, 'DisplayName', 'Objective Z = 0.7 m)')
xlabel('Sample number')
ylabel('Position (m)')
title('Drone X, Y, and Z position (Filtered + Objective Points')
legend("Location","northwest")
hold off

% I thought I would create a little legend in the console because there are a lot of figures and I couldn't figure out a way
% to label them anything other than a number.
fprintf("\nFigures 2, 3, 4, and 5 all corespond to Exercise 6 with the collection of drone data\n")

%xindex = 1;
%for i = 1:length(p_x)
%    if ( (p_x(i) >= -2) && (p_x(i) <= 2) )      I decided to keep this given that I worked with it for a long time to iterate
%        x_sorted(xindex) = p_x(i);              through the data set for the numbers I wanted, before realizing I can just use
%       xindex = xindex + 1;                     abs() trick like I did above. This approach is more in line with the other
%    end                                         excerises we did in the lab but takes up way more space and overcomplicates things.
%end
```

```
Figures 2, 3, 4, and 5 all corespond to Exercise 6 with the collection of drone data
```


Filtered X Position (within +-0.2m
X Position
Objective (0 m)
X Position (m)
Sample number
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
-0.2
0
500
1000
1500
2000
2500
3000


Filtered Y Position (within +-0.2m
Y Position
Objective (0 m)
Y Position (m)
Sample number
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
500
1000
1500
2000
2500
3000





Excercise 7

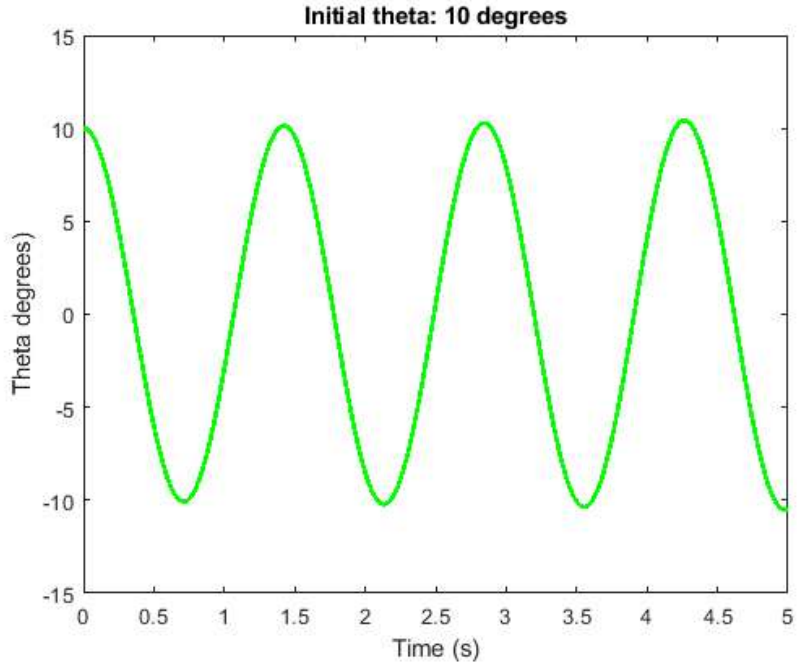
Plotting theta with respect time using discrete-time dynamics equations

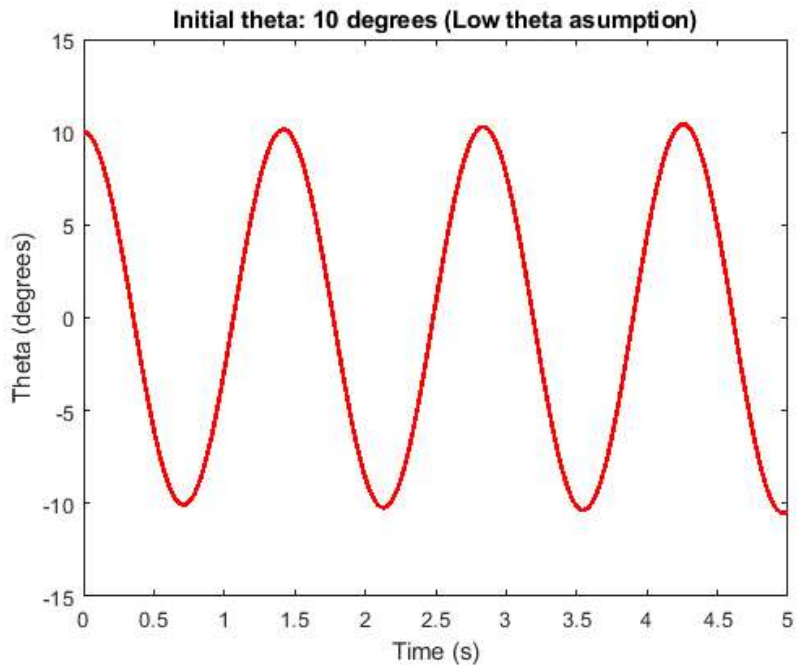
```
% ----------------------- Part D -----------------------
% Calling functions with a small value of theta

figure(6)
PlotThetaA(deg2rad(10), 0, 0.001, 5)
title('Initial theta: 10 degrees')
figure(7)
PlotThetaB(deg2rad(10), 0, 0.001, 5)
title('Initial theta: 10 degrees (Low theta asumption)')

% There is no noticable difference between the two graphs at a theta value
% this small. Doing the calculation by hand for thetaDot shows that because
% the value is so small, the discrepency between sin(theta) and theta itself
% (in radians, of course) is incredibly small, which leads to the very
% similar behavior seen on the graphs.
```

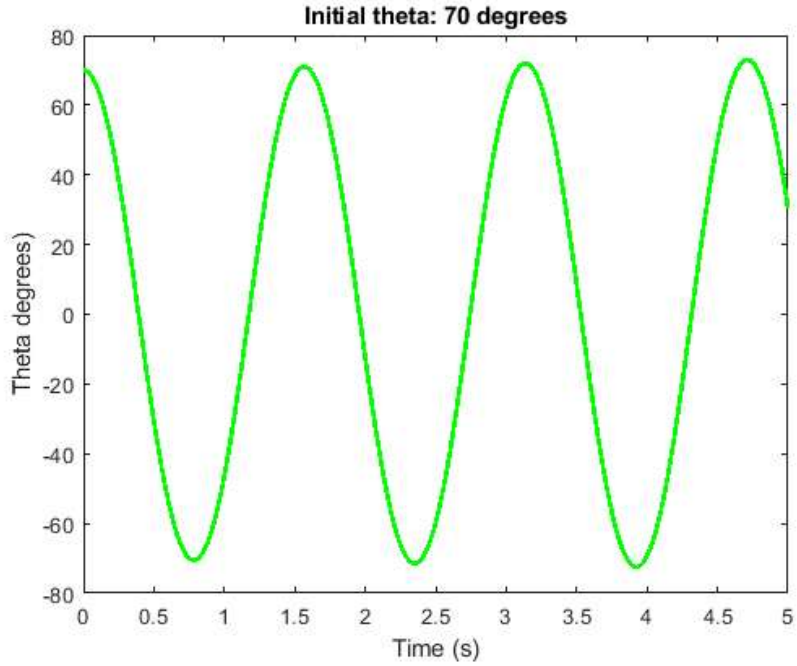
```
% ----------------------- Part E -----------------------
% Calling functions with a significantly larger value of theta

figure(8)
PlotThetaA(deg2rad(70), 0, 0.001, 5)
title('Initial theta: 70 degrees')
figure(9)
PlotThetaB(deg2rad(70), 0, 0.001, 5)
title('Initial theta: 70 degrees (Low theta asumption)')


% Here the difference is still not very obvious, but is definitely visible
% by just looking at the end point behavior. The function that assumes
% small values of theta ends at almost the complete opposite angle
% (-73 degrees vs ~40) as the function that correctly impliments changing
% values of thetaDot. Doing a manual calculation once again, the discrepency
% between sin(theta) and theta is too large to ignore, which is clear looking
% at the graphs.



% ----------------------- Part F -----------------------
% Calling functions with small values of theta but larger values of delta (t)

figure(10)
PlotThetaA(deg2rad(10), 0, 0.02, 5)
title('Initial theta: 10 degrees with larger delta (t)')
figure(11)
PlotThetaB(deg2rad(10), 0, 0.02, 5)
title('Initial theta: 10 degrees & larger delta (t) (Low theta asumption)')

% For this final test, the end behavior between the two graphs is almost
% exactly identical, very similar to the first test. This is expected,
% because after shrinking the value of delta (t), we give the two graphs
% less effective time to diverge (less total iterations through the loop).
% A manual calculation once again provides a very small discrepency between
% the two values of sin(theta) and theta.


% I added another little legend here in the console for the same reason as above, there are a ton of figures to keep track of
fprintf("Figures 6 through 11 are all the variations of plotting the angle theta\nfor a swinging pendulum using dynamics equations\n")

% ------------------------ Part C Functions -----------------------
% Takes theta, theta dot, the time step, and the total time as inputs and plots theta with respect to time for those values
function [] = PlotThetaA(theta0, thetaDot0, timeStep, totalTime)
    g = 9.81;
    L = 0.5;

    % Create time vector
    t = 0:timeStep:totalTime;
    N = length(t);

    % Initialize arrays with 0s
    theta = zeros(1, N);
    thetaDot = zeros(1, N);

    % Set up the initial conditions
    theta(1) = theta0;
    thetaDot(1) = thetaDot0;

    % Iterate over theta, incrementing theta and thetaDot according to dynamics equation
    for k = 1 : N - 1
        theta(k+1) = theta(k) + timeStep * thetaDot(k);
        thetaDot(k+1) = thetaDot(k) - timeStep * (g/L) * sin(theta(k));
    end

    % Now for the plot
    plot(t, theta * (180/pi), 'g', 'LineWidth', 2)
    xlabel('Time (s)')
```

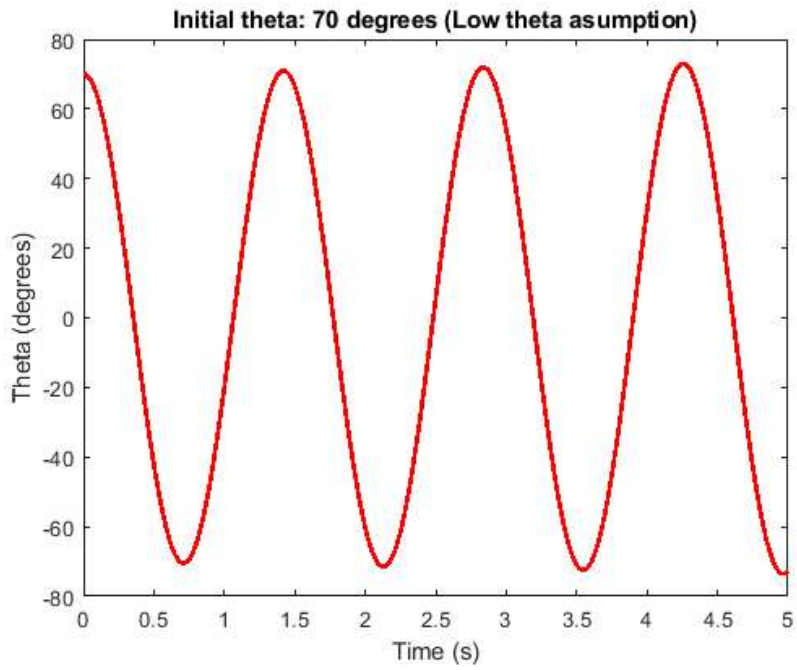
```
    ylabel('Theta degrees)')
    title("Pendulum Angle Theta Recorded Over Time")
end


% Do almost the exact same thing, except for the change to thetaDot over time
function [] = PlotThetaB(theta0, thetaDot0, timeStep, totalTime)
    g = 9.81;
    L = 0.5;

    % Create time vector
    t = 0:timeStep:totalTime;
    N = length(t);

    % Initialize arrays with 0s
    theta = zeros(1, N);
    thetaDot = zeros(1, N);

    % Set up the initial conditions
    theta(1) = theta0;
    thetaDot(1) = thetaDot0;

    % Iterate over theta, incrementing theta and thetaDot according to dynamics equation
    for k = 1 : N - 1
        theta(k+1) = theta(k) + timeStep * thetaDot(k);
        thetaDot(k+1) = thetaDot(k) - timeStep * (g/L) * theta(k);
    end

    % Now for the plot
    plot(t, theta * (180/pi), 'r', 'LineWidth', 2)
    xlabel('Time (s)')
    ylabel('Theta (degrees)')
    title("Pendulum Angle Theta Recorded Over Time (Assuming small Theta Values)")
end
```

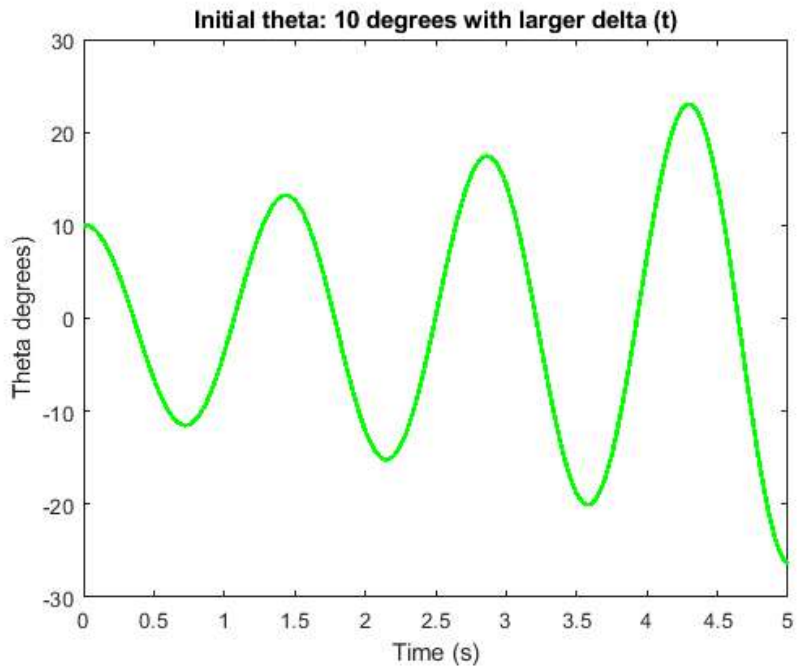
```
Figures 6 through 11 are all the variations of plotting the angle theta
for a swinging pendulum using dynamics equations
```

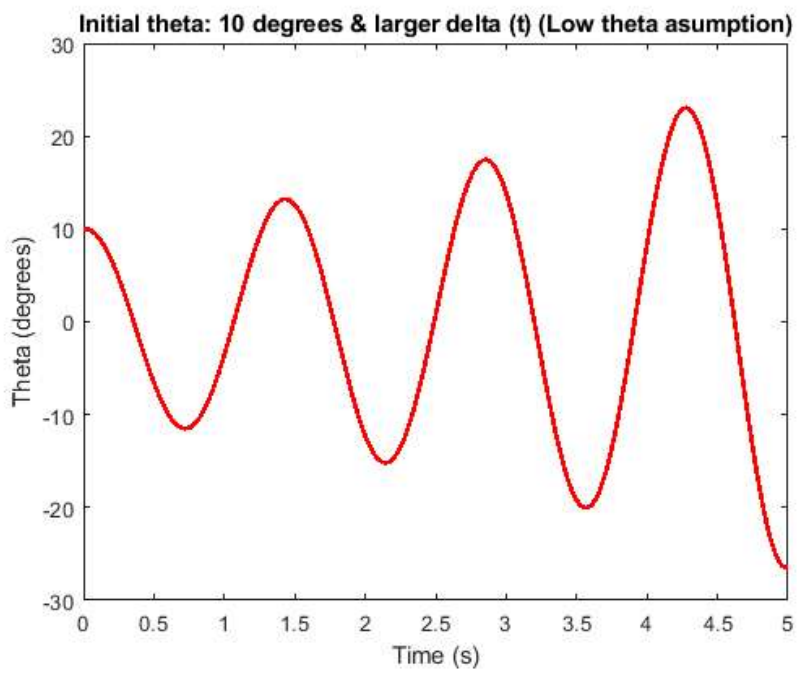

Initial theta: 10 degrees
Theta degrees)
Time (s)
15
10
5
0
-5
-10
-15
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5


Initial theta: 10 degrees (Low theta asumption)
Theta (degrees)
Time (s)
15
10
5
0
-5
-10
-15
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5


Initial theta: 70 degrees
Theta degrees)
Time (s)
80
60
40
20
0
-20
-40
-60
-80
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5


Initial theta: 70 degrees (Low theta asumption)
Theta (degrees)
Time (s)
80
60
40
20
0
-20
-40
-60
-80
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5





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